

Cardiovascular and respiratory responses to apneas with and without face immersion in exercising humans

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¹Department of Cell and Organism Biology, Lund University, SE-223 62 Lund; ²Department of Anesthesiology and Intensive Care, Lund University Hospital, SE-221 85 Lund; and ³Department of Natural and Environmental Sciences, Mid Sweden University, SE-851 70 Sundsvall, Sweden

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Andersson, Johan P. A., Mats H. Linér, Anne Fredsted, and Erika K. A. Schagatay. Cardiovascular and respiratory responses to apneas with and without face immersion in exercising humans. *J Appl Physiol* 96: 1005–1010, 2004. First published October 24, 2003; 10.1152/jappphysiol.01057.2002.—The effect of the diving response on alveolar gas exchange was investigated in 15 subjects. During steady-state exercise (80 W) on a cycle ergometer, the subjects performed 40-s apneas in air and 40-s apneas with face immersion in cold (10°C) water. Heart rate decreased and blood pressure increased during apneas, and the responses were augmented by face immersion. Oxygen uptake from the lungs decreased during apnea in air (–22% compared with eupneic control) and was further reduced during apnea with face immersion (–25% compared with eupneic control). The plasma lactate concentration increased from control (11%) after apnea in air and even more after apnea with face immersion (20%), suggesting an increased anaerobic metabolism during apneas. The lung oxygen store was depleted more slowly during apnea with face immersion because of the augmented diving response, probably including a decrease in cardiac output. Venous oxygen stores were probably reduced by the cardiovascular responses. The turnover times of these gas stores would have been prolonged, reducing their effect on the oxygen uptake in the lungs. Thus the human diving response has an oxygen-conserving effect.

diving reflex; bradycardia; vasoconstriction; oxygen and carbon dioxide stores; breath holding

THE HUMAN DIVING RESPONSE is mainly characterized by bradycardia, decreased cardiac output, peripheral vasoconstriction, and increased arterial blood pressure (8, 12, 13). Both the cardiac and vascular components of the human diving response are initiated by apnea and augmented by several factors, among which face immersion in cold water is especially recognized (13, 31). During apnea, the cardiac output decreases because of concurrent reductions in stroke volume and heart rate (5, 11, 39). Peripheral blood flow is reduced because of arterial vasoconstriction (14, 39), whereas blood flow in the carotid artery increases (18, 33), suggesting a redistribution of blood flow toward vital organs during apnea. The human diving response is also initiated by apnea and augmented by face immersion during exercise (3, 4, 37, 40).

Although the diving response of some diving animals has been shown to be O₂ conserving (6), the potential O₂-conserving effect of the human diving response has been questioned (22, 38, 39). However, recent observations indicate that the human diving response may also have an O₂-conserving effect. In three elite breath-hold divers, apneas and breath-hold dives

were associated with blood lactate accumulation and a reduced O₂ uptake that was not observed during apneas in untrained control subjects (9, 10). A more pronounced diving response resulted in less arterial O₂ desaturation, in both resting and exercising subjects (1, 2, 25). All of these results support the view that the human diving response has an O₂-conserving effect.

The alveolar gas exchange during apnea has previously been studied extensively (e.g., Refs. 16, 21, 23, 28, 29, 39). However, no study has shown whether the apneic alveolar gas exchange can be attenuated by face immersion, which has been shown for the arterial O₂ desaturation (1, 2). The effects of the diving response on the alveolar gas exchange can be studied by eliciting responses of different magnitudes in the same individuals under otherwise identical experimental conditions. Therefore, to elucidate a possible conserving effect of the human diving response on the pulmonary O₂ store, we recorded cardiovascular responses and alveolar gas exchange during apneas with and without face immersion in cold water in exercising men. It was hypothesized that the pulmonary O₂ uptake would be reduced and plasma lactate concentration increased during apneas compared with eupneic control. Furthermore, we hypothesized that a more pronounced diving response during apneas with face immersion would be associated with a lower pulmonary O₂ uptake and higher plasma lactate concentration compared with during apneas in air.

METHODS

The research ethics committee at Lund University had approved the experimental protocol. After receiving a description of the procedures and an explanation of potential risks involved, all subjects gave their informed consent.

Subjects. A group of 15 healthy, male subjects volunteered for the study. Their mean age was 25 yr (range: 19–33 yr), height was 186 cm (179–195 cm), weight was 85 kg (72–108 kg), vital capacity was 6.3 liters (5.0–7.1 liters), and residual volume was 1.9 liters (1.4–2.5 liters). They were active breath-hold divers (10 subjects), training breath-hold diving at least 2 h/wk to daily, Class B according to Schagatay and Andersson (36), or recruited among subjects having performed long apneas in previous studies at our laboratory (5 subjects; Classes C and D). In addition to their breath-hold diving training, their physical training averaged 6 h/wk (0–11 h/wk). All subjects were nonsmokers, and they reported to the laboratory after at least 2 h without any heavy meal or caffeine-containing beverages.

Experimental protocol. First, after local infiltration with <1 ml of lidocaine (Xylocaine, 10 mg/ml, Astra, Södertälje, Sweden), an arterial catheter was inserted in the radial artery at the right wrist. A

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venous catheter was inserted in a superficial vein in the right elbow. Then, the subject mounted a cycle ergometer (Monark 829 E, Monark Exercise, Varberg, Sweden), and the probes of the noninvasive instruments were attached, after which the vital capacity was measured. A water container used for face immersion was positioned on a shelf in front of the ergometer. The subject's arms could rest on the shelf on both sides of the container. When stable cardiovascular values were observed, recordings began. After an additional 2 min of rest on the cycle ergometer, the subject performed upright, steady-state, dynamic leg exercise at a workload of 80 W for ~50 min.

During the exercise, the subject performed a total of eight apneas alternating between apnea with face immersion and apnea with the face held above the water surface. The starting order alternated between apnea with face immersion and apnea in air among the subjects. During apneas with face immersion, the entire face was immersed with a maintained body position by just flexing the neck. Also, during the apneas in air, the neck was flexed, to maintain the body in virtually the same position during all apneas. Exercise began 5 min before the first apnea and continued until 5 min after the last apnea. The breath-holding time was kept at 40 s in both conditions, and apneas were spaced by 5 min. The water temperature was maintained between 10 and 11°C, and the ambient air temperature was 22–26°C.

Before each apnea, the subject exhaled to his residual volume through an open-circuit spirometry mouthpiece and inhaled, from a rubber bag, a volume of air equal to 80% of the individual sitting vital capacity. On command from the experimenter, the face was lifted and the apnea terminated after 40 s with a maximal expiration through the mouthpiece. After the exercise, the residual volume was determined with a nitrogen-dilution technique (35).

Measurements. Before the test, an electrocardiogram was recorded and checked for anomalies (Cardisun 501, Fukuda ME Kogoyo, Tokyo, Japan). The vital capacity was measured with a spirometer (Micro Plus, Micro Medical, Rochester, UK). During the experiment, the heart rate was continuously recorded with a heart rate monitor (Polar Vantage, Polar Electro Oy, Kempele, Finland). The arterial blood pressure was continuously recorded with a photoplethysmometer with the cuff on the left middle finger (Finapres 2300, Ohmeda, Madison, WI). It has previously been reported that it is possible to accurately record changes in mean arterial blood pressure with the Finapres during both exercise and apnea (17, 34). The left hand was positioned at the same level relative to the heart throughout the whole experiment. A relative index of skin capillary blood flow in the left thumb was continuously recorded in 10 of the subjects with a laser-Doppler flowmeter (Advanced Laser Flowmeter 21, Advance, Tokyo, Japan). Arterial hemoglobin O₂ saturation (SaO₂) was continuously recorded with an earlobe pulse oximeter (Biox 3700e, Ohmeda). Respiratory flow and volume as well as expiratory O₂ and CO₂ fractions for determination of alveolar gas exchanges were recorded with an open-circuit spirometry system (CPX/D Cardiopulmonary Exercise System, Medical Graphics, Minneapolis, MN). Temperature, barometric pressure, and humidity were measured in the laboratory just before each experimental session. All of the cardiovascular and respiratory parameters were recorded from 2 min before exercise until 2 min after exercise.

Arterial and venous blood samples for blood-gas analysis were collected 1 min before apnea and simultaneously with the expiration ending apnea. In two of the subjects, it was not possible to collect any venous samples, and the blood gas and pH results are, therefore, based on 13 subjects. Blood gases and pH were immediately analyzed with a blood-gas analyzer (AVL Compact 3, AVL List, Graz, Austria). Arterial blood samples for analysis of the arterial plasma lactate concentration were collected 1 min before and 1 min after each apnea in all subjects. Blood samples were kept on ice and cold centrifuged within 60 min. The separated plasma was then stored at –18°C until it was analyzed at the Laboratory of Clinical Chemistry at Lund

University Hospital (Modular P, Roche Diagnostics/Hitachi Instruments).

Data analysis. Average values for heart rate, mean arterial blood pressure, skin blood flow, SaO₂, and alveolar gas exchanges were calculated for the period 90–30 s before each apnea and used as control. Apneic values for heart rate, mean arterial blood pressure, and skin blood flow were calculated as average values for the last 10 s of each apnea. For SaO₂, an average value was calculated for the 10-s period ending with the nadir SaO₂ value. The relative change from control for these apneic 10-s periods was calculated. As an estimate of the myocardial O₂ demand (32), the rate-pressure product (heart rate × systolic arterial blood pressure: dimensionless parameter) during the control and apneic periods was calculated.

The O₂ and CO₂ exchange between the lungs and blood and the respiratory exchange ratio (RER) during apneas were calculated from the differences between volumes of O₂ and CO₂ in the lungs at the beginning of and at the end of apnea. The volume of O₂ in the lungs at the beginning of apnea was calculated by adding the volume of O₂ in the rubber bag to the volume of O₂ in the residual volume (using the end-tidal fraction of O₂ in the last, maximal expiration before each apnea). The same calculations were done for the volumes of CO₂ and inert gases. For the determination of the lung volume at the end of apnea, it was assumed that the volume of inert gases in the lungs was constant during apnea (16). The O₂ and CO₂ volumes in the lungs at the end of apnea were thus calculated by using the end-apnea lung volume and the end-tidal fractions in the maximal expiration after each apnea. The time-averaged alveolar gas exchange during apneas was calculated as the change in volumes of O₂ and CO₂ in the lungs during apneas divided by the corresponding breath-holding time, including the time required for inhalation from and exhalation to residual volume.

For each subject, individual mean values from the three last apneas of each condition were calculated for all parameters. From the 15 individual means, a group mean ± SE was calculated. The apneic values were compared with eupneic control, and the apnea in air was compared with apnea with face immersion by using paired *t*-test. The level used for accepting significance was *P* < 0.05.

RESULTS

All reported data are group means ± SE. The eupneic control values for the cardiovascular and respiratory parameters during steady-state exercise did not differ before apnea in air and apnea with face immersion, except the rate-pressure product, which was higher before apnea with face immersion (Table 1).

Table 1. Eupneic steady-state values of the cardiovascular and respiratory parameters

	Pre-A	Pre-AFI
Heart rate, beats/min	99 ± 3	99 ± 3
MAP, mmHg	115 ± 4	117 ± 4
Rate-pressure product	16,537 ± 1,025	17,011 ± 1,096*
SaO ₂ , %	98 ± 0.1	98 ± 0.1
PET _{O₂} , Torr	98 ± 1	98 ± 1
PET _{CO₂} , Torr	43 ± 1	44 ± 1
O ₂ uptake, ml/min STPD	1,277 ± 19	1,272 ± 20
CO ₂ elimination, ml/min STPD	1,221 ± 18	1,222 ± 19
RER	0.96 ± 0.01	0.96 ± 0.01
[p-La] _a , mmol/l	1.1 ± 0.1	1.1 ± 0.1

Values are means ± SE from 15 subjects from the control period, 90–30 s before apnea in air (Pre-A); and apnea with face immersion (Pre-AFI). MAP, mean arterial blood pressure; rate-pressure product, heart rate × systolic arterial blood pressure; SaO₂, arterial hemoglobin O₂ saturation; PET_{O₂}, end-tidal PO₂; PET_{CO₂}, end-tidal PCO₂; RER, respiratory exchange ratio; [p-La]_a, arterial plasma lactate concentration. **P* < 0.05 between Pre-A and Pre-AFI.

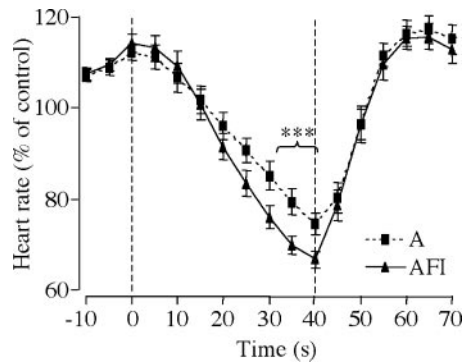


Fig. 1. Relative change from control in heart rate (HR) for each 5-s period before, during, and after apnea in air (A) and apnea with face immersion (AFI). Vertical dashed lines indicate the beginning and end of apneas. Values are means $\pm$ SE from 15 subjects. *** P < 0.001 between A and AFI for the average relative change during the last 10 s of apnea.

The diving response, noticeable as a heart rate reduction from the control level, was initiated by apneas and augmented by face immersion (Fig. 1). The skin blood flow was also reduced from control during apneas, whereas the mean arterial blood pressure increased during apneas (Fig. 2). The increase in mean arterial blood pressure was augmented during apnea with face immersion compared with during apneas in air. With the changes in heart rate and arterial blood pressure, the rate-pressure product did not change during apnea in air but was reduced from the control level during apnea with face immersion (Fig. 2).

The SaO_2 was reduced during apneas. From the pulse oximeter recordings, it could be calculated that the decrease from the control level during the last 10 s before the nadir SaO_2 value was attenuated during apnea with face immersion, $6 \pm 1\%$, compared with during apnea in air, $8 \pm 1\%$ (P < 0.001). Because of an averaging algorithm by the pulse oximeter, the nadir pulse oximeter values after apneas were higher (apnea in air: $88 \pm 1\%$; apnea with face immersion: $90 \pm 1\%$) than the SaO_2 determined in the end-apneic arterial blood samples (apnea in air: $80 \pm 2\%$; apnea with face immersion: $84 \pm 1\%$).

The alveolar PO_2 , as represented by end-tidal PO_2 (PETO_2), was reduced from the preapnea level during apneas. In the last, maximal expiration just before apneas, the PETO_2 did not differ before apnea in air and apnea with face immersion, 98 ± 2 vs. 99 ± 2 Torr. After apnea in air the PETO_2 was lower, 50 ± 1

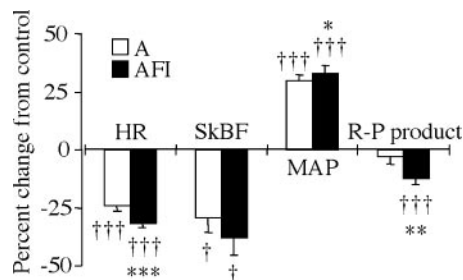


Fig. 2. Percent change from control in HR, skin blood flow (SkBF), mean arterial blood pressure (MAP), and rate-pressure product (R-P product) during A and AFI. Values are means $\pm$ SE from 15 subjects (SkBF, 10 subjects). $\dagger P$ < 0.05 and $\dagger\dagger\dagger P$ < 0.001 for absolute apneic value compared with respective control value. * P < 0.05, ** P < 0.01, and *** P < 0.001 between A and AFI.

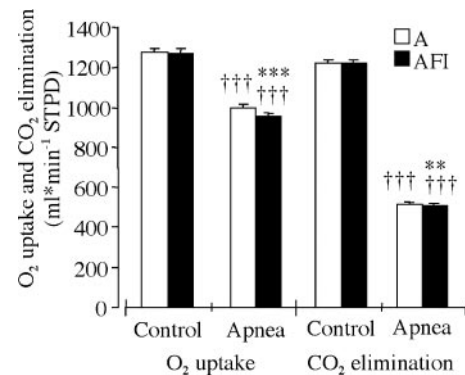


Fig. 3. Time-averaged lung-to-blood O_2 uptake and blood-to-lung CO_2 elimination during eupneic control and during A and AFI. Values are means $\pm$ SE from 15 subjects. $\dagger\dagger\dagger P$ < 0.001 compared with respective control value. ** P < 0.01 and *** P < 0.001 between A and AFI.

Torr, than after apnea with face immersion, 53 ± 1 Torr (P < 0.001). The O_2 uptake from the lungs to the blood during the apneic period was reduced compared with the eupneic control level during both apnea in air and apnea with face immersion (Fig. 3). During apnea in air, the O_2 uptake fell to $78 \pm 1\%$ of the eupneic control level and during apnea with face immersion to $75 \pm 1\%$ (P < 0.001, air vs. face immersion).

The alveolar PCO_2 increased during apneas. In the last, maximal expiration just before apneas, the end-tidal PCO_2 (PETCO_2) did not differ before apnea in air and apnea with face immersion, 45 ± 1 vs. 44 ± 1 Torr. The PETCO_2 was higher after apnea in air, 62 ± 1 Torr, than after apnea with face immersion, 61 ± 1 Torr (P < 0.001). The blood-to-lung CO_2 elimination during apneas was reduced compared with the eupneic control level (Fig. 3). During apnea in air and apnea with face immersion, the CO_2 elimination fell to $42 \pm 1\%$ and $41 \pm 1\%$ of the control level, respectively. With the changes in CO_2 elimination and O_2 uptake, the RER was reduced during apneas compared with the eupneic control level, to 0.52 ± 0.01 and 0.53 ± 0.01 during apnea in air and apnea with face immersion, respectively (P < 0.001, air vs. face immersion).

During the first 15 s of the recovery from apneas, the O_2 uptake fell markedly (Fig. 4). The initial fall was followed by a transient increase in O_2 uptake. This increase in O_2 uptake lasted for ~ 30 s, after which the O_2 uptake slowly returned to the control level. There was no difference in O_2 uptake, calculated as the total volume of O_2 uptake during the period 15–195 s after apnea in air and apnea with face immersion. The

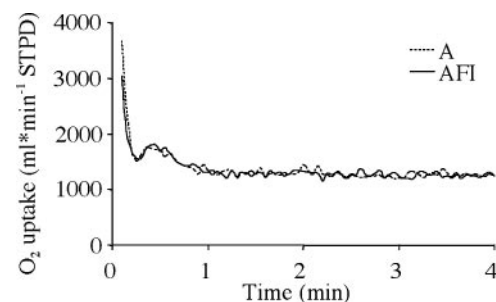


Fig. 4. Time course of the O_2 uptake after A and AFI. Values are means from 15 subjects.

Table 2. P_{O_2} in peripheral arterial and venous blood 1 min before apneas and at end of apneas

	Arterial		Venous	
	Before	End	Before	End
A	91 ± 2	53 ± 2‡	46 ± 4	43 ± 2
AFI	92 ± 2	58 ± 2*‡	47 ± 4	42 ± 2†

Values are means ± SE from 13 subjects given in Torr. A, apnea in air; AFI, apnea with face immersion. † $P < 0.05$ and ‡ $P < 0.001$ compared with before apneas. * $P < 0.001$ compared with End-A.

CO_2 elimination followed a similar, although slower, pattern of changes in the early recovery period.

The arterial and venous P_{O_2} , P_{CO_2} , and pH at 1 min before and at the end of apneas are shown in Tables 2–4. Notable is that the arterial P_{O_2} is higher and arterial P_{CO_2} lower at the end of apnea with face immersion compared with at the end of apnea in air, and that the venous P_{CO_2} is lower than the arterial P_{CO_2} at the end of apneas ($P < 0.001$).

The arterial plasma lactate concentration after apneas was higher compared with the concentration before apneas (Fig. 5). The increase in plasma lactate concentration was augmented during apnea with face immersion compared with during apnea in air.

DISCUSSION

In the present study with steady-state exercise, the alveolar gas exchange was reduced from eupneic control during apnea in air, confirming earlier observations (9, 23, 29). The alveolar gas exchange was further reduced during apnea with cold-water face immersion, showing that the lung O_2 store is depleted at a slower rate during apnea with face immersion. This is probably due to the augmented diving response and associated decrease in cardiac output. The increase in plasma lactate concentration after apneas indicates that the reduced O_2 uptake is associated with a shift toward more anaerobic metabolism during apnea. Thus we confirm earlier suggestions that the human diving response has an O_2 -conserving effect during apnea, similar to the diving response of some diving mammals.

It has previously been demonstrated that the diving response is augmented by cold-water face immersion, in both resting and exercising humans (4, 19, 39, 40), and our results are in accordance with these observations. The cardiac output is reduced in relation to the degree of the heart rate reduction during apnea, and there is a concurrent increase in systemic vascular resistance, which reduces peripheral blood flow (5, 24, 39). Furthermore, during apnea with a lung volume above the functional residual capacity and with relaxed respiratory

Table 3. P_{CO_2} in peripheral arterial and venous blood 1 min before apneas and at end of apneas

	Arterial		Venous	
	Before	End	Before	End
A	40 ± 1	56 ± 1‡	45 ± 1	47 ± 1†
AFI	41 ± 1	54 ± 1*‡	46 ± 1	48 ± 1†

Values are means ± SE from 13 subjects given in Torr. † $P < 0.05$ and ‡ $P < 0.001$ compared with before apneas. * $P < 0.01$ compared with End-A.

Table 4. pH in peripheral arterial and venous blood 1 min before apneas and at end of apneas

	Arterial		Venous	
	Before	End	Before	End
A	7.37 ± 0.01	7.29 ± 0.01‡	7.35 ± 0.01	7.34 ± 0.01†
AFI	7.37 ± 0.01	7.30 ± 0.01*‡	7.35 ± 0.01	7.34 ± 0.01†

Values are means ± SE from 13 subjects. † $P < 0.01$ and ‡ $P < 0.001$ compared with before apneas. * $P < 0.01$ compared with End-A.

muscles, the increased intrathoracic pressure reduces the venous return and, consequently, stroke volume and cardiac output (11, 27). This latter effect on cardiac output is most likely identical, however, in the two experimental conditions of this study. Cardiac output would, therefore, be lower during apnea with face immersion than during apnea in air because of the more pronounced heart rate reduction. The increase in mean arterial blood pressure in the present study supports earlier observations indicating that there is a pronounced arterial vasoconstriction during both apnea in air and apnea with face immersion during exercise and that the vasoconstriction is more pronounced during apnea with face immersion (39).

The myocardial O_2 demand, estimated by the rate-pressure product, was reduced during apnea with face immersion, as observed earlier by Bjertnaes et al. (5). Although this is an indirect measure of the myocardial workload, the reduction is noteworthy, because it has been claimed that the increasing arterial blood pressure during apneas would offset any O_2 -conserving effects of the bradycardia, because of an increase in cardiac afterload (15, 22).

Because the cardiac output is reduced during apnea, the lung-to-blood O_2 uptake will be reduced as a result of a reduced pulmonary blood flow, i.e., changes in alveolar gas exchange are largely determined by changes in cardiac output (23, 26, 27, 29). Therefore, the reduced O_2 uptake and consequent higher alveolar and arterial P_{O_2} and SA_{O_2} during apnea with face immersion could partly be a result of an augmented decrease in cardiac output during apnea with face immersion compared with during apnea in air. In addition to reducing O_2 uptake in the lungs, a decrease in cardiac output will reduce

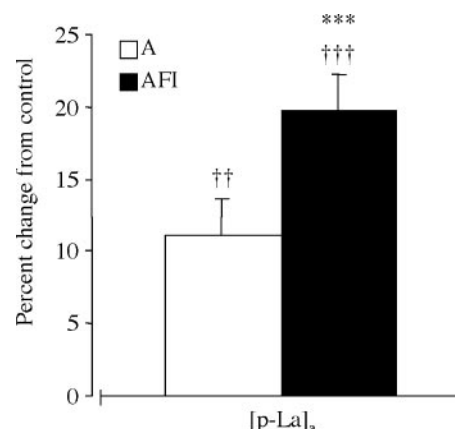


Fig. 5. Increase in arterial plasma lactate concentration ($[p-La]_a$) in samples collected 1 min after compared with 1 min before apnea in A and AFI. Values are means ± SE from 15 subjects. †† $P < 0.01$ and ††† $P < 0.001$ for absolute value compared with respective control value. *** $P < 0.001$ between A and AFI.

venous O₂ stores if the tissue O₂ consumption is maintained (7, 26, 27, 29). However, the reduced cardiac output will simultaneously slow the flow of deeply desaturated venous blood to the lungs, i.e., prolong the turnover time of the venous O₂ stores, thereby reducing their influence on the pulmonary O₂ uptake. The diving response will in this way affect the utilization of lung and tissue O₂ stores during apneas, and the lung O₂ store will be preserved at the expense of the venous O₂ stores.

The redistributions of blood flow and venous blood volume during apneas will further reduce the O₂ uptake and the venous O₂ stores. The peripheral vasoconstriction will increase the arterial-to-venous difference in O₂ content in the peripheral tissues (22). Furthermore, venous blood is displaced peripherally because of high intrathoracic pressure. The result is a reduced O₂ content of the blood draining into the increased peripheral venous blood pool (29, 30). Simultaneously, the turnover time for these peripheral venous O₂ stores will be prolonged by the peripheral vasoconstriction and peripheralization of venous blood volume (26). Hence, the reduction in venous O₂ stores contributes to the observed reduction in O₂ uptake from the lungs. With the decrease in cardiac output and the peripheral vasoconstriction during apnea, the lung O₂ store will be preserved for use by mainly the heart and the brain. Therefore, with a pronounced diving response, the lung O₂ store will be preserved for a longer period, thereby delaying the time of development of hypoxic levels that threatens the integrity of the heart and the brain during apnea. Taken together, the results thus indicate that the human diving response has an O₂-conserving effect during apnea.

The transient increase in O₂ uptake after 15 s of breathing in the recovery period after apneas supports the conclusion of reduced peripheral venous O₂ stores. It is suggested that this increase reflects the circulatory delay of the peripheral venous O₂ stores. When more desaturated venous blood enters the lung circulation after some 15–45 s of resumed breathing, the O₂ uptake is increased.

It should be emphasized that, in addition to just reducing the lung-to-blood O₂ uptake, the human diving response could also affect the aerobic rate of peripheral tissues by limiting the O₂ delivery. If the peripheral O₂ delivery is reduced, affected tissues are restricted to derive energy from aerobic metabolism with tissue O₂ stores or from anaerobic energy production with lactic acid formation and breakdown of high-energy phosphates (9, 10). We provide additional support for these observations because we also found an increase in arterial plasma lactate concentration after apneas compared with the concentration before apneas. It is noteworthy that this occurred at a workload that is normally not associated with lactic acid accumulation during eupnea. Furthermore, the increase was greater during apnea with face immersion, when the diving response was augmented. This increase in arterial lactate concentration could be influenced by a number of factors. First, the samples were collected from the radial artery. The plasma lactate concentration would thus probably have been higher if the samples had been collected from the femoral vein. Second, we only analyzed the lactate concentration in blood samples collected 1 min before and 1 min after apneas. A higher peak in arterial plasma lactate concentration after apneas might have been present, before or after the collection of the postapnea samples. Furthermore, the plasma lactate concentration is not a direct measure of the lactic acid formation but rather the

summation of formation, release, and clearance. Nevertheless, the increase in plasma lactate concentration supports the view of an increased anaerobic metabolic rate during apneas, correlated to the magnitude of the diving response.

When normal CO₂ elimination from the body is blocked during apnea, the P_{CO₂} values of the lungs and tissues increase (16). The blood-to-lung CO₂ elimination is continuing during the beginning of apnea. Later during the apnea, CO₂ delivery to the lungs almost ceases and may even be reversed (16, 21, 23, 28). The reversal in CO₂ exchange is explained by an increase in alveolar and arterial P_{CO₂} above that of the mixed-venous P_{CO₂}, caused by lung volume shrinkage. This, in turn, is attributed to the continuous lung-to-blood O₂ uptake (16, 21). During apneas of relatively short durations, as in the present study, the lung and blood CO₂ stores will be of major importance for storing the CO₂ produced by the tissues (9). The reduced RER during apneas compared with eupneic control in the present study supports the idea that CO₂ is stored in the blood and tissues during apneas. Also, it is reasonable to assume that a portion of the CO₂ will be stored in the muscles where it is produced, because of reduced blood flow during apnea. Actually, the time course of CO₂ elimination during the recovery period, which had a similar increase after some 15 s of breathing just as the O₂ uptake, supports this assumption. The lower CO₂ elimination during apnea with face immersion compared with during apnea in air may be a result of an augmented decrease in cardiac output and prolonged turnover times for tissue CO₂ stores in the former condition.

The reduced P_{CO₂} of the venous blood from the arm compared with that of the arterial blood at the end of apneas is of interest. Even considering the Haldane effect, this marked fall in P_{CO₂} can be calculated (20) to correspond to a decrease in blood CO₂ content from 52–53 ml/100 ml in the arterial blood to 49–50 ml/100 ml in the venous blood. This indicates that some CO₂ is stored in the resting arm.

In conclusion, the human diving response is confirmed to have an O₂-conserving effect, probably through reductions in cardiac output and peripheral blood flow during apnea. The apneic lung-to-blood O₂ uptake is reduced compared with eupneic control, because of the circulatory adjustments, thereby preserving the lung O₂ store for vital organs during apnea. In addition, increased plasma lactate concentration after apnea indicates that the anaerobic metabolic rate may increase as a consequence of reduced peripheral O₂ delivery because of the diving response. The changes observed during apnea in air were augmented during apnea with face immersion, further substantiating the conclusions about the O₂-conserving effect.

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